

DMRG in era of MPS.

→ what is MPS?

→ what is DMRG?

→ how do they come together?

for N-body wavefunction, $\dim(\text{Hilbert space } H) = d^N$ ($d \rightarrow 2$)
 → not computationally efficient!

Assumption: low energy, low entanglement states of Hilbert space can be represented as a **MPS**

$$|\psi\rangle = \sum A^i \dots A^n |i \dots n\rangle$$

Why low entanglement? GS of local Hamiltonian is less entangled than random state. → AREA LAW

Local Hamiltonian $S_A \propto L^{d-1}$
 Random $S_A \propto L^d$ } surface vs volume scaling
 → only boundary of

Notations

c $c \rightarrow$ scalar

$$\frac{A}{i,j}$$

$A_{ij} \rightarrow$ value

each branch is another index.

v_i $v_i \rightarrow$ vector

$$\frac{A}{i,j}$$

$A_{ij}^c \rightarrow$ tensors...

Sample contraction calculations:

$$\sum_{j,k} A_{ijm} B_{jkn} = T_{ilm}$$

$\Rightarrow$



Complexity:

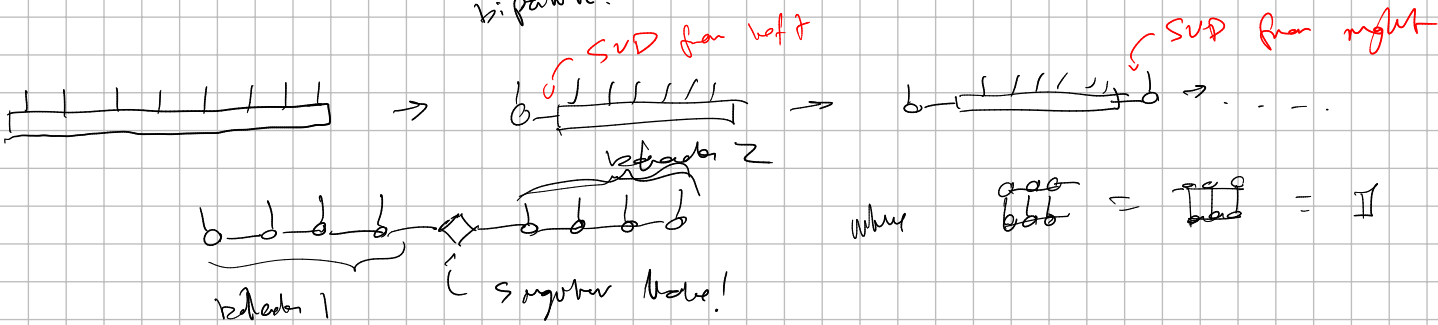
$\prod_{\text{conn.}} D_i \cdot \prod_{\text{open}} D_j \Rightarrow$ product of dimensionality of each "leg"

How do we get MPS from wavefunction? SVD!

→ Using SVD, keep decomposing until we get Schmidt decomposition.

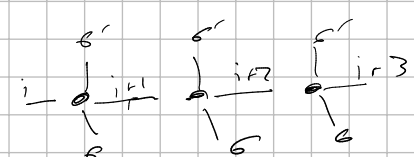
$$|\psi\rangle = \sum \psi_{ij} |i\rangle |j\rangle \xrightarrow{\text{SVD}} \sum \lambda_{ij} |i\rangle |j\rangle$$

is pairwise.

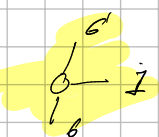


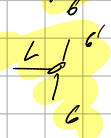
If wavefunctions become MPS, Hamiltonian become MPOs

MPO:

$$W_{i, in}^{a_1, a'_1} W_{in, ir1}^{a_2, a'_2} W_{ir1, ir2}^{a_3, a'_3} \dots \Rightarrow$$


We need to define behavior for 3 cases:

$$W_{\uparrow}^{a, a'} =$$


$$W_{\downarrow}^{a, a'} =$$


$$W_{i, ir1}^{a, a'} =$$
